MSG No. 57.389 P_Abcm [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[0, \frac{1}{4}, 0\right]$	`[1,2,11,12]`
2	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[3,4,9,10]`
3	$\left[0, \frac{3}{4}, 0\right]$	`[5,6,15,16]`
4	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[7,8,13,14]`

Table 2: Wyckoff site: `4b`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[1,2,11,12]`
2	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,4,9,10]`
3	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[5,6,15,16]`
4	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	`[7,8,13,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `..2'/m`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[1,8,12,13]`
2	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	`[2,7,11,14]`
3	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	`[3,6,10,15]`
4	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[4,5,9,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `..2'/m`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,8,12,13]`
2	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,7,11,14]`
3	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,6,10,15]`
4	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[4,5,9,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,11,15]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,6,12,16]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,7,9,13]`
4	$\left[0, 0, \frac{1}{2}\right]$	`[4,8,10,14]`

Table 6: Wyckoff site: `4f`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,5,11,15]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,6,12,16]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,7,9,13]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,8,10,14]`

Table 7: Wyckoff site: `4g`, site symmetry: `2'm'm`

No.	position	mapping
1	$\left[x, 0, \frac{3}{4}\right]$	`[1,8,10,15]`
2	$\left[x, \frac{1}{2}, \frac{1}{4}\right]$	`[2,7,9,16]`
3	$\left[-x, \frac{1}{2}, \frac{3}{4}\right]$	`[3,6,12,13]`
4	$\left[-x, 0, \frac{1}{4}\right]$	`[4,5,11,14]`

Table 8: Wyckoff site: `8h`, site symmetry: `..2'`

No.	position	mapping
1	$\left[0, \frac{1}{4}, z\right]$	`[1,12]`
2	$\left[0, \frac{1}{4}, -z\right]$	`[2,11]`
3	$\left[0, \frac{3}{4}, \frac{1}{2} - z\right]$	`[3,10]`
4	$\left[0, \frac{3}{4}, z + \frac{1}{2}\right]$	`[4,9]`
5	$\left[0, \frac{3}{4}, -z\right]$	`[5,16]`
6	$\left[0, \frac{3}{4}, z\right]$	`[6,15]`
7	$\left[0, \frac{1}{4}, z + \frac{1}{2}\right]$	`[7,14]`
8	$\left[0, \frac{1}{4}, \frac{1}{2} - z\right]$	`[8,13]`

Table 9: Wyckoff site: `8i`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{4}, z\right]$	`[1,12]`
2	$\left[\frac{1}{2}, \frac{1}{4}, -z\right]$	`[2,11]`
3	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2} - z\right]$	`[3,10]`
4	$\left[\frac{1}{2}, \frac{3}{4}, z + \frac{1}{2}\right]$	`[4,9]`
5	$\left[\frac{1}{2}, \frac{3}{4}, -z\right]$	`[5,16]`
6	$\left[\frac{1}{2}, \frac{3}{4}, z\right]$	`[6,15]`
7	$\left[\frac{1}{2}, \frac{1}{4}, z + \frac{1}{2}\right]$	`[7,14]`
8	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2} - z\right]$	`[8,13]`

Table 10: Wyckoff site: `8j`, site symmetry: `.2'.`

No.	position	mapping
1	$[0, y, 0]$	`[1,11]`
2	$\left[0, \frac{1}{2} - y, 0\right]$	`[2,12]`

continued ...

Table 10

No.	position	mapping
3	$\left[0,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[3,9]
4	$\left[0,\, -y,\, \frac{1}{2}\right]$	[4,10]
5	$[0,\, -y,\, 0]$	[5,15]
6	$\left[0,\, y+\frac{1}{2},\, 0\right]$	[6,16]
7	$\left[0,\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[7,13]
8	$\left[0,\, y,\, \frac{1}{2}\right]$	[8,14]

Table 11: Wyckoff site: 8k, site symmetry: .2'.

No.	position	mapping
1	$\left[\frac{1}{2},\, y,\, 0\right]$	[1,11]
2	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, 0\right]$	[2,12]
3	$\left[\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[3,9]
4	$\left[\frac{1}{2},\, -y,\, \frac{1}{2}\right]$	[4,10]
5	$\left[\frac{1}{2},\, -y,\, 0\right]$	[5,15]
6	$\left[\frac{1}{2},\, y+\frac{1}{2},\, 0\right]$	[6,16]
7	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[7,13]
8	$\left[\frac{1}{2},\, y,\, \frac{1}{2}\right]$	[8,14]

Table 12: Wyckoff site: 8l, site symmetry: 2..

No.	position	mapping
1	$\left[x,\, \frac{1}{4},\, 0\right]$	[1,2]
2	$\left[-x,\, \frac{3}{4},\, \frac{1}{2}\right]$	[3,4]
3	$\left[-x,\, \frac{3}{4},\, 0\right]$	[5,6]
4	$\left[x,\, \frac{1}{4},\, \frac{1}{2}\right]$	[7,8]
5	$\left[x,\, \frac{3}{4},\, \frac{1}{2}\right]$	[9,10]
6	$\left[-x,\, \frac{1}{4},\, 0\right]$	[11,12]
7	$\left[-x,\, \frac{1}{4},\, \frac{1}{2}\right]$	[13,14]
8	$\left[x,\, \frac{3}{4},\, 0\right]$	[15,16]

Table 13: Wyckoff site: 8m, site symmetry: ..m

No.	position	mapping
1	$\left[x,\, y,\, \frac{3}{4}\right]$	[1,8]
2	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	[2,7]
3	$\left[-x,\, y+\frac{1}{2},\, \frac{3}{4}\right]$	[3,6]
4	$\left[-x,\, -y,\, \frac{1}{4}\right]$	[4,5]
5	$\left[x,\, y+\frac{1}{2},\, \frac{1}{4}\right]$	[9,16]
6	$\left[x,\, -y,\, \frac{3}{4}\right]$	[10,15]
7	$\left[-x,\, y,\, \frac{1}{4}\right]$	[11,14]
8	$\left[-x,\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	[12,13]

Table 14: Wyckoff site: `8n`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,15]`
2	$\left[x,\ \frac{1}{2},\ -z\right]$	`[2,16]`
3	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[3,13]`
4	$\left[-x,\ 0,\ z+\frac{1}{2}\right]$	`[4,14]`
5	$[-x,\ 0,\ -z]$	`[5,11]`
6	$\left[-x,\ \frac{1}{2},\ z\right]$	`[6,12]`
7	$\left[x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,9]`
8	$\left[x,\ 0,\ \frac{1}{2}-z\right]$	`[8,10]`

Table 15: Wyckoff site: `16o`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	`[2]`
3	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[3]`
4	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[4]`
5	$[-x,\ -y,\ -z]$	`[5]`
6	$\left[-x,\ y+\frac{1}{2},\ z\right]$	`[6]`
7	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[7]`
8	$\left[x,\ y,\ \frac{1}{2}-z\right]$	`[8]`
9	$\left[x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9]`
10	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	`[10]`
11	$[-x,\ y,\ -z]$	`[11]`
12	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	`[12]`
13	$\left[-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[13]`
14	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	`[14]`
15	$[x,\ -y,\ z]$	`[15]`
16	$\left[x,\ y+\frac{1}{2},\ -z\right]$	`[16]`